BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 096 · Lesson 096 · Little Man Computer: Instructions and the Accumulator

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the ACC for?

Q2. What does STA X do?

Q3. What does INP do?

Q4. Half adder A1,B1: what Sum and Carry?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LMC — Little Man Computer, simplified stored-program model.

mailbox — Memory location.

ACC — Accumulator for working results.

STA — Store ACC into memory.

LDA — Load ACC from memory.

New learning

Observe — LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

Apply — INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

Evaluate — OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

Worked example

Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?

1. Follow the reasoning — FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.

We Do · Practise with guidance

Modify the addition program to output second input minus first input; inputs 4 then 9.

Follow the reasoning

After first input store FIRST=4; second input ACC9; SUB FIRST → ACC5; OUT5.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace program above for inputs 4,7: ACC after each INP, STA and ADD; output. [6]

Question 2

Write an LMC program that reads one integer, stores it as N, then loads and outputs N. [6]

Question 3

Compare LDA FIRST, STA FIRST and ADD FIRST in relation to ACC and mailbox; give distinct effects. [6]

Exit Ticket · Answer before checking

STA FIRST overwrites ACC with the mailbox's value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Stores current/intermediate calculation.

Q2. Stores current ACC into mailbox X.

Q3. Reads input into ACC, replacing its former value.

Q4. Sum0 (XOR); Carry1 (AND).

You Do pages · Independent questions

Q1. ACC4 after first INP [1], FIRST4 after STA [1], ACC7 after second INP [1], ACC11 after ADD [2], output11 [1].

Q2. INP [1], STA N [1], LDA N [1], OUT [1], HLT [1], N DAT 0 [1].

Q3. LDA copies mailbox to ACC [2], STA copies ACC to mailbox [2], ADD adds mailbox value into ACC [2].

Exit Ticket · Model answer

False. LDA does that; STA writes to the mailbox.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: